FRA Diagnostic Analysis Report

Standard: IEEE C57.149-2012 / IEC 60076-18 Standards-Aligned SFRA Diagnostic Pipeline | **System:** AI-Powered Hybrid Transformer Integrity Monitor

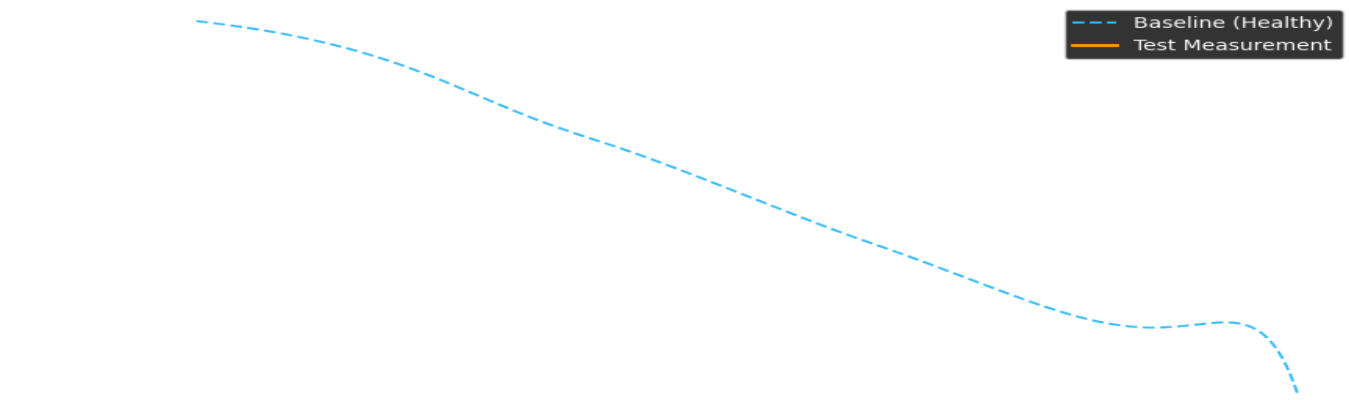
Asset / Transformer ID: TX-100	**Inspection Date:** 2025-04-10
Winding Configuration: Tertiary	**Overall Status / Severity: Warning** (Medium)

Overall Health Integrity Score Blended Physics (60%) + ML (40%)	**12.8%**	**Condition Assessment:** Critical Fault Condition **AI Fault Signature:** Insulation Degradation (32.0% conf.) **Overall Correlation (CCF):** 0.0000 | **Max Dev:** 0.00 dB

Sub-band Diagnostic Breakdown (IEEE C57.149 Standards Table)

Low (Core)	< 2 kHz	N/A	N/A	**Insufficient Data**
Mid (Winding)	2 - 100 kHz	N/A	N/A	**Insufficient Data**
High (Insulation)	> 100 kHz	N/A	N/A	**Insufficient Data**

Frequency Response Bode Plot Comparison



Actionable Recommendations & Maintenance Guidance

Primary Assessment: Minor deviation detected. Schedule a DGA (Dissolved Gas Analysis) to confirm internal state.

Targeted Protocol: Moderate deviations observed in frequency response signature. Schedule oil sampling for Dissolved Gas Analysis (DGA) within 30 days. Re-sweep SFRA at next scheduled maintenance outage to monitor fault progression.

Component-Level Guidance (IEEE C57.149):
- **Core (Low Band, < 2 kHz):** Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.
- **Winding (Mid Band, 2–100 kHz):** Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.
- **Insulation (High Band, > 100 kHz):** Insufficient data points (< 10 points) in this sub-band. Re-sweep frequency response.

Deterministic Physics Floor: As mandated by IEEE C57.149-2012 and IEC 60076-18, the diagnostic status is governed strictly by the worst-performing sub-band (Low: Core, Mid: Winding, High: Insulation). Machine Learning predictions provide fault signature classification and escalation without overriding physical limits.